Chapter 2: Basic Topology

Section: Connected Sets

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Separated and Connected Sets

Connected Subsets of the Real Line

Proof of Theorem 2.47

Summary

Why This Section Matters

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- One more key property: whether a set comes in one piece
- Goal: make "one piece" precise $\implies$ connectedness

Separated and Connected Sets

Definition 2.45: Separated Sets

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Two subsets A and B of a metric space X are separated if

$$A \cap \bar{B} = \varnothing \quad \text{and} \quad \bar{A} \cap B = \varnothing.$$

That is, no point of A lies in the closure of B, and no point of B lies in the closure of A.

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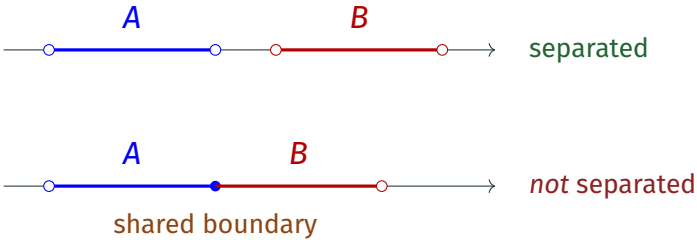
Definition 2.45 — Connected Set

A set $E \subset X$ is connected if E is *not* a union of two nonempty separated sets.

Picture: Separated Sets



Separation = gap in both directions: neither set touches the other's closure.

Remark 2.46: Disjoint Does Not Mean Separated

Remark 2.46 — Key Fact

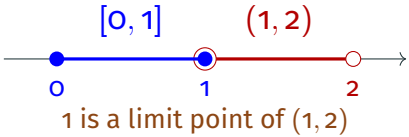
- Separated sets are always disjoint.
- Disjoint sets need not be separated.

Remark 2.46: Disjoint but *Not* Separated

Example — $[0,1]$ and $(1,2)$

Disjoint, but $1 \in [0,1]$ and 1 is a limit point of $(1,2)$, so

$$[0,1] \cap \overline{(1,2)} = \{1\} \neq \varnothing.$$



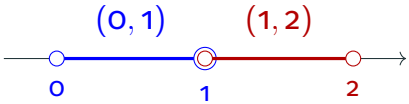
Remark 2.46: *Truly* Separated

Example — $(0,1)$ and $(1,2)$

$$\overline{(0,1)} = [0,1], \quad \overline{(1,2)} = [1,2].$$

$$(0,1) \cap [1,2] = \varnothing, \qquad [0,1] \cap (1,2) = \varnothing.$$

Hence $(0,1)$ and $(1,2)$ are separated.



Connected Subsets of the Real Line

Theorem 2.47: Connected Subsets of $\mathbb{R}^1$

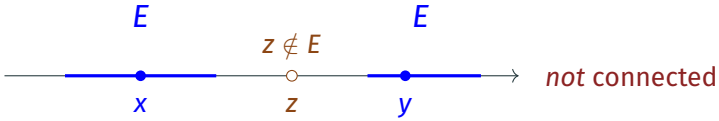
Theorem 2.47

A subset E of the real line $\mathbb{R}^1$ is connected if and only if it has the following property:

$$\boxed{\text{If } x, y \in E \text{ and } x < z < y, \text{ then } z \in E.}$$

In plain words: E contains every number sitting between two of its own points.

Picture: The “No Holes” Property



A gap between two points of E immediately breaks connectedness.

Proof of Theorem 2.47

Proof Strategy: Theorem 2.47

Goal: Prove the biconditional E connected $\iff$ E has the in-between property.

We prove both directions by contrapositive:

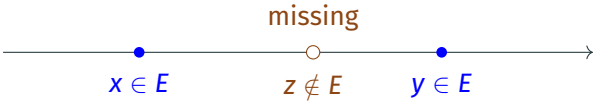
(A) If the property *fails*, then E is *not* connected.
$\rightsquigarrow$ construct an explicit separation of E.

(B) If E is *not* connected, then the property *fails*.
$\rightsquigarrow$ use a separation $E = A \cup B$ and take $z = \sup(A \cap [x, y])$.

Part (A): Property Fails $\implies$ Not Connected

Setup

Suppose $\exists\, x, y \in E$ and $z \in (x, y)$ with $z \notin E$.



Goal: exhibit $E = A_z \cup B_z$ with A_z, B_z nonempty and separated.

Part (A): Constructing the Split

Setup

Suppose $\exists\, x, y \in E$ and $z \in (x, y)$ with $z \notin E$.

Step 1 — Define A_z and B_z

$$A_z = E \cap (-\infty, z), \qquad B_z = E \cap (z, \infty).$$

Since $z \notin E$: $\quad E = A_z \cup B_z$.

Nonemptiness: $\quad x \in A_z$ (since $x < z$), $\quad y \in B_z$ (since $z < y$).

Part (A): Separation and Conclusion

Step 2 — A_z and B_z are separated

$$A_z \subset (-\infty, z) \implies \overline{A_z} \subset (-\infty, z],$$

$$B_z \subset (z, \infty) \implies \overline{B_z} \subset [z, \infty).$$

Only possible overlap: $\{z\}$. But $z \notin A_z$ and $z \notin B_z$, so

$$A_z \cap \overline{B_z} = \varnothing, \qquad \overline{A_z} \cap B_z = \varnothing.$$

Conclusion

$E = A_z \cup B_z$ with A_z, B_z nonempty and separated $\implies$ E is not connected.

Part (B): Not Connected $\implies$ Property Fails

Setup

Suppose E is *not* connected. Then

$$E = A \cup B, \quad A, B \neq \varnothing, \quad A, B \text{ separated}.$$

Pick $x \in A$, $y \in B$. Assume WLOG $x < y$.

Plan: find z^* with $x < z^* < y$ and $z^* \notin E$.

Part (B): The Key Construction — $z = \sup(A \cap [x, y])$

Step 1 — Define z

$$z = \sup\,(A \cap [x, y]).$$

Note: $A \cap [x, y]$ is nonempty (x is in it) and bounded above by y.

Recall: Theorem 2.28

If $E \subset \mathbb{R}$ is nonempty and bounded above, then $\sup E \in \bar{E}$.

Part (B): Locating z

Step 2 — Locate z

By Theorem 2.28, $z \in \bar{A}$. Since $\bar{A} \cap B = \varnothing$: $\quad z \notin B$.

Three bounds:

- $x \leq z$ (since $x \in A \cap [x, y]$),
- $z \leq y$ (since y is an upper bound),
- $z \neq y$ (since $y \in B$, $z \notin B$).

Hence $\boxed{x \leq z < y.}$

Part (B): Case 1 — $z \notin A$

Case 1: $z \notin A$

Already know: $z \notin B$. So $z \notin A \cup B = E$.

And $x \leq z < y$; in fact $x < z$ because $x \in A$ while $z \notin A$.

$$\Longrightarrow \quad \boxed{x < z < y \text{ and } z \notin E.}$$

The property $x, y \in E,\ x < z < y \Rightarrow z \in E$ fails at this z.

Part (B): Case 2 — $z \in A$

Case 2: $z \in A$

Separation: $A \cap \bar{B} = \varnothing$, so $z \notin \bar{B}$.

Hence some open neighborhood of z misses B entirely: $\exists z_1$ with

$$z < z_1 < y \quad \text{and} \quad z_1 \notin B.$$

But $z_1 > z = \sup(A \cap [x, y])$ and $z_1 < y$, so $z_1 \notin A$ either.

$\Longrightarrow \quad \boxed{x < z_1 < y \text{ and } z_1 \notin E.}$

In both cases the in-between property fails ■

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- Disjoint $\neq$ separated (2.46): $[0, 1]$ and $(1, 2)$ are disjoint but not separated.

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Key takeaways from this section:

- Separated sets (2.45): neither set meets the other's closure.
- Connected sets (2.45): cannot be split into two nonempty separated pieces.
- Disjoint $\neq$ separated (2.46): $[0, 1]$ and $(1, 2)$ are disjoint but not separated.
- Theorem 2.47: $E \subset \mathbb{R}^1$ connected $\iff$ E has the in-between property $\iff$ E is an interval (broad sense).

Coming up next (Chapter 3): *Numerical Sequences and Series* — convergence, Cauchy sequences, and the central limit notion of analysis.